

Professional Summary

State Engineer in Computer Science, graduated from the National Higher School of Computer Science (ESI Algiers) with a specialization in Computer Systems and Software. Solid expertise in software architecture, high-availability distributed systems design, data interoperability, and security. Trilingual (Arabic, French, fluent English), capable of collaborating with international and technical partners. Proven track record in end-to-end complex platform development (Backend, Mobile, Web, DevOps) and integration of advanced AI/IoT solutions (Edge Computing, RAG, Federated Learning).

Education

State Engineer’s Degree in Computer Science, Systems and Software 2021 – 2026
National Higher School of Computer Science (ESI) Algiers, Algeria
High School Diploma (Scientific Track), Grade: 18.34/20 (Highest Honors) 2021

Professional Experience

Software Engineer August 2025 – August 2026
KABAS AI, deep-tech startup, AI and software solutions for industrial clients Algiers, Algeria

- HSE Inspection Platform:** Designed and developed end-to-end. *Offline-first* Flutter mobile application (QR-code inspection points, local sync queue), FastAPI/PostgreSQL backend with real-time WebSockets, and a React analytics dashboard. Integrated JWT security, containerized deployment with Docker, and Jenkins CI/CD pipelines.
- Dalile - Sovereign Multi-Agent AI Platform:** Developed the React frontend interface for KABAS AI’s industrial document intelligence system. Integrated a multi-agent architecture (RAG, scientific search, analytics), workspace management with dynamic PDF citation highlighting, live indexing status, and DMS/Drive connectors.

Final Year Project: Distributed LLM-IoT & Data Security Sep 2025 – June 2026
National Higher School of Computer Science (ESI), defended June 28, 2026 Algiers, Algeria

- Designed a highly secure embedded system for mental health: local processing of IoT sensor data into structured reports and execution of an on-device Edge LLM without cloud data leaks. Integrated decentralized and federated learning to guarantee absolute data privacy.

Research Internship: Data Analysis & Natural Language Processing Jul 2024 – Sep 2024
National Higher School of Computer Science (ESI) Algiers, Algeria

- Designed feature descriptors for detecting relationship conflicts on social networks. Handled data cleaning, structured an annotated corpus, and built an analytics dashboard interface (Django, MySQL, Vue.js).

Academic Projects & Technological Achievements

IRCHAD: Embedded System & Computer Vision for the Visually Impaired Nov 2024 – Jan 2025
National Higher School of Computer Science (ESI) Algiers, Algeria

- Developed the onboard embedded computer: obstacle detection and positioning using computer vision (visual tags), streaming real-time data to a mobile app via WebSockets for voice and haptic guidance.

Smart SMS Gateway & Developer API (IoT/ESP32) March 2025
Personal / Academic Project Algiers, Algeria

- Developed an SMS gateway on an ESP32/GSM microcontroller. Designed a REST API, Python/JS SDKs, Web/Mobile interfaces, and webhook integrations (Zapier, Google Calendar).

Technical Skills

Languages Python, JavaScript/TypeScript, Dart, Kotlin, PHP, C/C++
Backend & APIs FastAPI, Django, Flask, Node.js/Express, REST APIs, WebSockets, Socket.IO
Front & Mobile React, TailwindCSS, Ant Design, shadcn/Radix UI, Flutter (BLoC), Kotlin
Data & Search PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
AI & Architecture Multi-Agent RAG, LLM APIs, Fine-tuning (QLoRA/LoRA), System Architecture
IoT & Embedded ESP32, Raspberry Pi, Arduino IDE, IoT Sensors, Computer Vision
DevOps Docker, Docker Compose, Jenkins CI/CD, Git/GitHub

Languages

Arabic: Native | French: Fluent | English: Fluent (Technical & Professional)